

Project: Analysis of Alberta's Electricity Demand and Price

Requirements Analysis

Prices of electricity in Alberta fluctuate constantly making it challenging for grid operators, energy producers, and large consumers to identify when they will need to provide, purchase, and use electricity. This can be a problem, I will be looking to see if there are patterns in electricity demand and price that can help operators, producers, and consumers.

I will use historical hourly data from the Alberta Electric System Operator. The dataset includes Alberta Internal Load which represents demand and price. The goal of analyzing this data will be to identify trends in electricity demand and see how they correlate with price fluctuations. I will aim to answer what, if any, are the daily, weekly, and seasonal patterns in demand and if they exist how does demand correlate with the pool price?

System Modeling

I will use a simple data processing pipeline model

Data Flow:

1. Load the hourly metered volumes and pool prices into a panda's data frame inside a jupyter notebook.
2. Clean the data for relevant columns
3. Calculate a 7-day running average for the demand and price columns.
4. Generate plots for hourly demand vs the 7-day running average and the hourly price vs the 7-day running average.

Requirements Engineering

Functional requirements:

- The system must load the AESO CSV file
- The system must calculate a 7-day running average for demand
- The system must calculate a 7-day running average for the price
- The system must generate a image containing the two subplots
- The final graphic must display at least three data series.

Non-functional requirements:

- The running average calculations must be arithmetically correct
- The script must load and process two years of hourly data in under 10 seconds
- The final plots must include a title, axis labels, and a legend for all data series displayed

Software Evolution

This project could be expanded over time by incorporating weather data to see its correlation with demand and price. The visualizations could be integrated into an interactive web-based data dashboard that could allow users to select different date ranges and running average windows.

Citations:

All AESO data used came from <https://www.aeso.ca/market/market-and-system-reporting/data-requests/hourly-generation-metered-volumes-and-pool-price-and-ail-data-2001-to-july-2025>

.gitignore template from <https://github.com/github/gitignore/blob/main/Python.gitignore>

Inspiration for using Panda's, Matplotlib, and Seaborn came from my experience with data analysis and visualization while working at Orpyx Medical Technologies.